$$8C_4 = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 70$$

[To find $10\,C_3$: In the numerator, first write the product of 3 natural numbers starting from 10 in the descending order and in the denominator write the factorial 3 and then simplify].

Compare $10C_8$ and $10C_2$

$$10\,C_8 = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3}{4 \times 3 \times 2 \times 1} = \frac{10 \times 9}{2 \times 1} = 45$$

$$10\,C_2 = \frac{10 \times 9}{2 \times 1} = 45$$

From the above we find $10\,C_8 = 10\,C_2$

Therefore this is also written as $10\,C_8 = 10\,C_{(10-8)} = 10\,C_2$

This is very useful, when the difference between n and r is very high in $n\,C_r$.

This property of combination is given as a result, $n\,C_r = n\,C_{n-r}$

To find $200\,C_{198}$, we can use the above formula as follows:

$$200\,C_{198} = 200\,C_{200-198}$$
$$= 200\,C_2$$
$$= \frac{200 \times 199}{2 \times 1}$$
$$= 19900.$$

Example 7.9

Out of 13 players, 11 Players are to be selected for a cricket team. In how many ways can this be done?

Solution:

Out of 13 Players, 11 Players are selected in $13C_{11}$ ways

i.e., $13C_{11} = 13C_2 = \frac{13 \times 12}{2 \times 1} = 78$

Example 7.10

In how many ways a committee of 5 members can be selected from 6 men and 5 women, consisting of 2 men and 3 women?

Solution:

For a committee, 2 men and 3 women members are to be selected. From 6 men, 2 men are selected in $6C_2$ ways. From 5 women, 3 women are selected in $5C_3$ ways.